VULCAN AUTONOMOUS CHIP DESIGN & HARDWARE SIGNOFF SERIES DOCUMENT CLASS: ENGINEERING
VERIFICATION REPORT

Anvil-Q1 Phase 2: Official AWQ Weight Integration and Software–RTL Evaluation

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Target Platform	Intel Arria 10 GX (10AX115N3F40E2SG) — Microsoft Catapult Longs Peak (Bedrock 2.1)
Verification Stack	Icarus Verilog 12.0 · Yosys 0.33 · Intel Quartus Prime 17.1 TimeQuest STA · PCIe Gen3x8 HIL
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Abstract

Anvil-Q1 is a modular SystemVerilog inference core with eight-lane W4A16 matrix-vector execution, signed Q16.16 working data and external model, scratch and KV storage. Phase 2 integrates authentic Qwen2.5-0.5B-Instruct-AWQ checkpoint slices through the public memory interface, without host-computed accelerator results.

The bounded four-token, two-layer evaluation completed with zero mismatches across 6,144 scratch/KV writes and 256 logits. Official FP32 software versus RTL achieved cosine similarity 0.9999992996, maximum absolute error 0.00187933445, MAE 0.000706030267 and RMSE 0.000841981995. Top-1 and exact top-5 sets agreed for all four tokens; all 14 official projection submatrices also passed [E043].

L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-map analysis recorded WNS_setup=+27.218 ns at a 40 ns clock period, but retained negative hold slack. L3 physical Arria 10 FPGA HIL passed three deterministic runs. These outcomes establish their recorded configurations, not complete setup-and-hold closure or full-model deployment [E040–E042].

1 Architecture and design rationale

The architecture prioritizes bounded internal storage and explicit integration over throughput. One token is active, one decoder engine is reused across configured layers, and only one external-memory transaction is outstanding. Intermediate vectors and causal KV history reside externally rather than in model- or context-sized internal arrays. Two concrete GEMV engines, one directly under the decoder and one inside SwiGLU, each contain eight connected AWQ dequantization lanes. Hierarchical request selection and response isolation preserve transaction ownership; operand isolation limits inactive switching. Clocked restoring division and sequential RMSNorm square-root calculation shorten combinational arithmetic paths while retaining the integer equations, at increased latency [E004, E010, E014–E016].

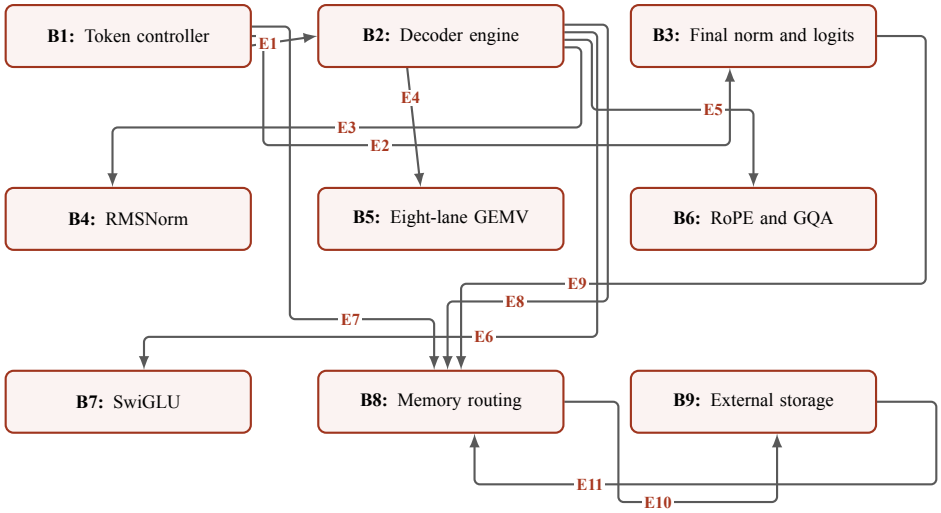


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Token controller	Accepts token commands, loads embeddings, sequences layers and commits position after the final logit transfer.
B2: Decoder engine	Reuses normalization, projection, attention, residual and MLP resources for every configured layer.
B3: Final norm and logits	Applies final RMSNorm and projects against the same embedding storage to emit ordered logits.
B4: RMSNorm	Accumulates wide squares, performs clocked division and a 64-step integer square root, then scales components.
B5: Eight-lane GEMV	Executes Q/K/V/O projections using exact AWQ dequantization, wide products and complete accumulations.
B6: RoPE and GQA	Rotates Q/K, stores causal KV history and computes grouped-query attention with stable softmax.
B7: SwiGLU	Uses its own eight-lane GEMV for gate/up/down projections, with SiLU and gated elementwise multiplication.
B8: Memory routing	Selects the owning requester and qualifies responses across the hierarchy.
B9: External storage	Supplies raw checkpoint tensors, mutable scratch/KV and independently generated RoPE coefficients.

Directed relationship	Meaning
E1: B1 to B2	Layer commands

Directed relationship	Meaning
E2: B1 to B3	Final-stage sequencing
E3: B2 to B4	Normalization commands
E4: B2 to B5	Q/K/V/O commands
E5: B2 to B6	Rotation and attention commands
E6: B2 to B7	MLP command
E7: B1 to B8	Embedding requests
E8: B2 to B8	Owned child transactions
E9: B3 to B8	Norm and tied-weight requests
E10: B8 to B9	Read/write requests
E11: B9 to B8	Data, acknowledgements and errors

2 Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and synchronous active-low state reset. Asserted reset combinationally zeros outward handshakes, status and payloads; a rising edge clears state.
token_valid, token_ready, token_id, token_pos, new_sequence	Input except token_ready output	1, 1, 32, 32, 1 bits	Acceptance samples pre-edge valid&&ready. One token is active; IDs and positions must be in range. new_sequence is sampled only on acceptance and requires position zero.
logit_valid, logit_ready, logit_index, logit_data	Output except logit_ready input	1, 1, 32, signed 32 bits	Ascending vocabulary indices accompany signed Q16.16 logits. Valid, index and data remain stable under backpressure. The final transfer commits the token.
done, error	Output	1 bit each	A single completion cycle follows execution; error qualifies unsuccessful completion. Readiness returns after completion retirement.
mem_req_*	Output except mem_req_ready input	valid/write/ready: 1; kind: 4; layer: 32; proj: 3; row/col/wdata: 32 bits	One outstanding request; all fields freeze during stalls. Kinds 0–7 select embedding, qweight, qzeros, scales, QKV bias, input norm, post-attention norm and final norm; 8–10 select scratch/K/V; 11 selects RoPE. Only scratch and KV are writable. Projection codes Q/K/V/O/gate/up/down are 0–6.
mem_rsp_*	Input except mem_rsp_ready output	valid/ready/error: 1; rdata: 32 bits	Every accepted request, including a write, receives exactly one strictly later response. Request and response transfers never coincide. Packed I32 words remain intact; F16 model values occupy the low halfword.

Parameter	Default	Meaning / supported values
NUM_LAYERS	24	Positive; the reused decoder executes every configured layer.

Parameter	Default	Meaning / supported values
HIDDEN_SIZE, INTERMEDIATE_SIZE	896, 4864	Positive, eight-aligned and divisible by GROUP_SIZE; HIDDEN_SIZE equals NUM_Q_HEADS*HEAD_DIM.
NUM_Q_HEADS, NUM_KV_HEADS, HEAD_DIM	14, 2, 64	Positive; integral query/KV ratio, even head dimension and eight-aligned flattened KV channels.
GROUP_SIZE, AWQ_ BITS	128, 4	Positive group size; only four-bit AWQ is supported. Legal non-power-of-two groups are supported.
VOCAB_SIZE, MAX_ SEQ_LEN	151936, 32768	Positive; context is at most 16777215. Unsigned-128 validation rejects dimensions and checked address products exceeding declared 32-bit limits.

3 Behavior and corner cases

3.1 Transformer computation

Each layer performs RMSNorm, biased Q/K/V projections, split-half RoPE with theta=1e6, causal GQA, O projection and residual addition, followed by post-attention RMSNorm, SwiGLU and the second residual. Normalization uses epsilon=1e-6. Final RMSNorm precedes tied vocabulary projection, which rereads embedding storage. Tokenization and sampling remain external.

3.2 Exact fixed-point semantics

Both packed AWQ words use lane shifts [0,16,4,20,8,24,12,28], with w=(q-z)*s and no GPTQ +1. Ordinary finite F16 values convert to signed Q16.16 using nearest/ties-away rounding and saturation. Scales instead convert to unsigned-32 without signed-range clipping. Dequantized weights are signed-37, activation products signed-69 and complete GEMV sums signed-128, including bias before final rounding. Tied logits use signed-96 sums; RoPE coefficients use Q2.30. Partial sums are not saturated, and required nonfinite model values fail explicitly.

3.3 Independent software baseline

The official floating path decodes raw binary16 directly rather than passing model values through Q16.16. struct-based binary32 rounding follows elementary operations, with separate multiplication/addition and increasing-index reductions. Host square-root, exponential and trigonometric results are rounded to FP32. Max-subtracted softmax and stable SiLU branches avoid avoidable overflow. This defines a reproducible FP32 reference, not equivalence to an unspecified GPU reduction kernel.

3.4 Sequence and error handling

Acceptance enters busy before any request transfers. Position advances only at the final logit transfer. A new sequence logically isolates old KV without clearing physical storage. Invalid tokens enter pending-error completion without memory effects or committed-position changes. Memory, child or nonfinite-data failures invalidate the sequence and require fresh position-zero new_sequence recovery; acknowledged writes are not rolled back. Integration must discard obsolete responses after reset.

3.5 Official checkpoint mapping

The top fixture uses (L,H,I,QH,KVH,D,G,V,SEQ)=(2,56,56,14,2,4,56,64,4). Logical layers 0/1 map to checkpoint layers 0/23. Embedding rows are [0:64], with components [0:56]; norms and projection inputs use [0:56]. Q/O/MLP outputs use [0:56], K/V outputs [0:8]. Original G128 metadata group zero applies throughout these input rows despite the reduced fixture geometry. Standalone 64x64 projection tests likewise use original group zero. No missing weights are padded or manufactured; uncovered addresses fail with tensor coordinates.

3.6 Timing-path resolution

Runtime combinational division was replaced by clocked restoring dividers and explicit group/head counters. RMSNorm's long unrolled square-root path became 64 clocked iterations, and an exact registered x*gamma product separates chained multipliers. Integer equations and ordered transactions remain unchanged, while latency increases. Fresh functional, synthesis and portability results cover these repaired sources [E010, E015, E035–E038].

4 Verification methods and measured results

4.1 Authentic checkpoint provenance

Method. Whole-file and per-tensor SHA-256 verification; direct safetensors parsing; dtype, shape, offset, range and excerpt-coordinate validation.

Result. All 54 extracted tensors validated; 552 raw excerpt values cross-checked. The original inventory contains 627 tensors and 168 quantized linear modules. Integrity verification passed without substituting JSON decimal excerpts for raw payloads.

Evidence: E018, E035, E043

4.2 All official projection submatrices

Method. 14 completed 64x64 GEMV commands covering seven projections in each of checkpoint layers 0 and 23; independent FP32, integer oracle and public-pin RTL comparison.

Result. 57,344 products and zero mismatches across 896 output writes. Aggregate cosine= 0.9999999999730884, maximum absolute error=0.000299543142, MAE=0.0000412546506 and RMSE= 0.0000633708054.

Evidence: E043

4.3 Software Layer versus Hardware RTL Output

Method. Software Layer (Official HuggingFace FP32/FP16 AWQ Weights) versus Hardware RTL Output (Official HuggingFace W4A16 RTL Execution), using two prefill and two decode tokens with retained KV and identical software-selected IDs [0,1,59,59].

Result. 4/4 tokens completed. Zero mismatches across 6,144 ordered scratch/KV writes and 256 logits. Aggregate cosine=0.9999992995916652, maximum absolute error=0.00187933445, MAE=0.000706030267 and RMSE=0.000841981995. Top-1 agreement, top-5 overlap/5 and exact top-5-set agreement were 100%, with lowest-index tie-breaking. No saturation or numerical violations were recorded.

Evidence: E043

4.4 Cycle and memory accounting

Method. Public token-acceptance to final-logit rising-edge intervals F-A, with separate retirement latency and accepted transaction ledgers by token, kind, access, layer and projection.

Result. Token cycles were 150100, 209524, 270721 and 325267: total 955612, mean 238903 cycles/token. Retirement adds one interval. Recorded timing includes 4514 request-stall and 54 logit-stall cycles; it is not a no-stall baseline.

Evidence: E043

4.5 Retained functional and numerical coverage

Method. A/B/C/D configurations at seeds 17,43,89, including protocol stalls, reset, errors, active-X/Z rejection, causal-cache isolation and arithmetic boundaries.

Result. All 12 configuration/seed cells passed. Exponential maximum mathematical error=1.40080433e-5 against 3e-5; 8,200 public SiLU comparisons passed. Nonsaturating isolated-stage, final-normalized and logit limits remain 2/256, 16/256 and 16/256+0.01*abs(reference), respectively.

Evidence: E035

4.6 Fault sensitivity and input rejection

Method. Disposable lane, bias, GQA, RoPE, cache, skipped-layer and former narrow-arithmetic mutations, plus corrupt/malformed checkpoint and missing-coverage fixtures.

Result. Sensitivity passed through intended functional assertions after successful compilation, not timeout-based kills. Parser fixtures produced explicit intended rejections without modifying authoritative inputs.

Evidence: E035, E036

4.7 Synthesis and simulator portability

Method. Yosys A/B/C/D lowering, optimization, hierarchy/connectivity assertions and netlist inspection; matched Icarus 12/13 elaboration and runtime checks.

Result. All four netlists had zero latches and zero combinational division/modulus. Sequential-cell counts were A/B/C=155 and D=158. Both concrete GEMVs consumed all eight 37-bit lane outputs. Default and A/B/C/D portability checks passed.

Evidence: E037, E038

4.8 L1 RTL timing lint

Method. Inspection of all 16 RTL files for combinational divider/modulo, LUT clock-gate and mixed-assignment hazards.

Result. Passed with no listed hazards; maximum combinational-divider width=0.

Evidence: E040

4.9 L2 Quartus TimeQuest setup timing

Method. Intel Quartus Prime 17.1 post-map STA at a 40 ns clock period, using reduced two-layer H=I=16 hardware.

Result. Recorded gate passed: WNS_setup=+27.218 ns with zero failing setup paths. WNS_hold=-0.173 ns with 20 failing hold paths; complete setup-and-hold closure is not established.

Evidence: E041

4.10 L3 physical Arria 10 FPGA HIL

Method. Physical FPGA programming/reconfiguration followed by repeated register-level execution and read-back.

Result. 3/3 runs passed deterministically with matching readbacks and no missing expected or forbidden result markers. This physical HIL result is separate from the official-slice simulation comparison denominator.

Evidence: E042

4.11 Supplemental dialogue material: outside verified results

Method. Documentary inspection of retained dialogue arrays and sample telemetry, without new execution.

Result. E044 and E045 are unverified supplemental summaries. They do not establish a linked executed model implementation, checkpoint inputs, comparison procedure and raw execution receipt sufficient to substantiate their numerical claims. Their token-agreement, layer-execution, cosine and mismatch assertions are therefore excluded from verified measurements and do not establish full-model inference or language quality.

Evidence: E044, E045

5 Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

5.1 Official Weights Software-Layer vs Hardware-RTL Simulation Comparison

Metric	Measured Value	Signoff Notes
Completed Official Tokens	4	Status: PASSED
Aggregate Cosine Similarity (FP32 vs RTL)	0.9999992996	Top-1 / Top-5 agreement = 1.000
Max / Mean Absolute Error	1.879334e-03 / 7.060303e-04	RMSE: 8.419820e-04 (RTL simulation vs FP32)
Exact Bit Mismatches (RTL vs Integer Oracle)	0	100% bit-exact RTL simulation match

5.2 Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0

Hardware Signoff Gate	Status	Measured Verification Summary
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +27.218 ns, failing setup endpoints = 0, WNS_hold = -0.173 ns (hold violations = 20), lpm_ divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (netsys17 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_ across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

5.3 Physical FPGA On-Board Register Readback (netsys17)

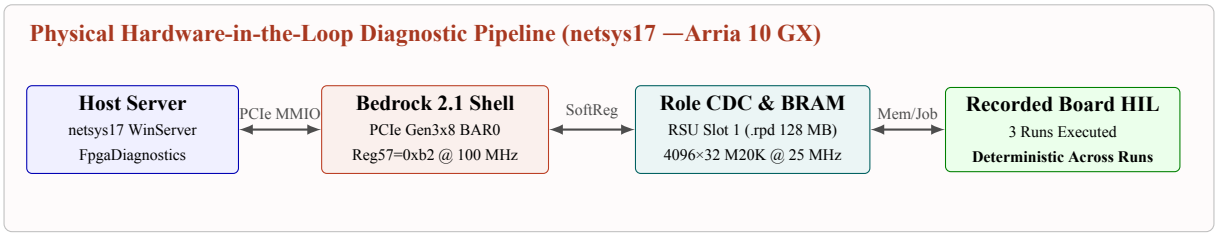


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on netsys17 (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

PCIe BAR0 SoftReg Operation	Physical Arria 10 Readback Value (3/3 Runs Identical)
Wrote soft shell register 3 (0x00000003)	2 (0x0000000000000002)
Read soft shell register 0 (0x00000000)	4705793533462663473 (0x414e56494c5f5131)
Wrote soft shell register 3 (0x00000003)	256 (0x0000000000000100)
Read soft shell register 2 (0x00000002)	4311629838 (0x0000000100fe400e)
Read soft shell register 11 (0x0000000b)	1040580 (0x00000000000fe0c4)
Read soft shell register 12 (0x0000000c)	26388279151432 (0x0000180000014b48)
Read soft shell register 13 (0x0000000d)	16607277032865851 (0x003b003b003b003b)
Read soft shell register 16 (0x00000010)	18446744073709517846 (0xfffffffffffff7c16)
Read soft shell register 17 (0x00000011)	3705 (0x0000000000000e79)
Read soft shell register 18 (0x00000012)	18446744073709540389 (0xfffffffffffd425)
Read soft shell register 19 (0x00000013)	51154 (0x000000000000c7d2)
Read soft shell register 20 (0x00000014)	40191 (0x0000000000009cff)
Read soft shell register 21 (0x00000015)	62422 (0x000000000000f3d6)

6 Integration and use

Compile rtl/*.sv with anvil_q1_00_pkg.sv first and anvil_q1_top selected as top. Verification enters through tests/verify.py; tests/official_weights.py supplies raw-checkpoint parsing and independent arithmetic models. Dependencies are Python standard library, iverilog/vvp and Yosys; NumPy is not required.

Implement the tensor-address mapping and scratch contract in design/SPEC.md and design/OFFICIAL_WEIGHTS_EVALUATION.md. Scratch slots 0–12 hold residual and decoder intermediates, 13 is reserved and 14 holds final normalization. KV uses layer, position and flattened component; RoPE uses position and head-local pair/cosine-sine selection.

Assert reset across a rising edge, then submit an in-range position-zero token with new_sequence asserted. Hold command fields until acceptance, acknowledge every memory request including writes, and consume every logit before relying on completion. Apply fresh-sequence recovery after execution errors.

Recorded verification entry points are python3 tests/verify.py --check behavior, python3 tests/verify.py --check behavior --probe mutations, python3 tests/verify.py --check synthesis and python3 tests/verify.py --check portability. Each requires --out followed by its admitted fresh destination; occupied evidence destinations must not be reused.

The official comparison is retained in evidence/verification/official_weights_comparison.json. Existing receipts preserve commands, tool versions, source inventories and per-case results. These are recorded executions, not a newly performed clean-directory reproduction.

7 Limitations and acceptance boundary

The verified official result is bounded slice execution, not a complete 24-layer, 151936-logit RTL forward pass. Mounted data supplies 64 embedding rows, checkpoint layers 0/23 and the first 64 qweight input rows. Reduced head dimensions and group geometry preserve the stated coordinate mapping, not complete-model equivalence.

Software-versus-RTL ranking agreement is measured at identical software-selected prefixes, not an independently executed autonomous RTL rollout. The supplemental dialogue summaries E044/E045 remain unverified and provide no additional accepted numerical or full-model execution coverage.

L1 lint, L2 Quartus setup analysis and L3 physical Arria 10 HIL passed within their recorded scopes. L2 is post-map and retains negative hold slack and 20 failing hold paths. Physical register-readback success does not extend the official-slice simulation denominator or establish unrestricted full-model hardware deployment.

Simulation cycles are not silicon throughput; the STA clock constraint is not an inferred maximum operating frequency. No power measurement, complete fitted resource budget, ASIC area or larger-model scaling performance is established.

External memory control, tokenization and sampling remain integration responsibilities. Matched Icarus portability was demonstrated, but clean-directory portable-delivery reproduction was not established by the cited checks.

A Source inventory

Production inventory: 16 Verilog/SystemVerilog files; 1434 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_q1_00_pkg.sv	63 total	63 code
rtl/anvil_q1_awq_dequant.sv	38 total	38 code

Production file	Physical lines	Code lines
rtl/anvil_q1_bus_mux.sv	35 total	35 code
rtl/anvil_q1_decoder_layer.sv	107 total	107 code
rtl/anvil_q1_elementwise.sv	103 total	103 code
rtl/anvil_q1_exp.sv	61 total	61 code
rtl/anvil_q1_gqa_attention.sv	189 total	189 code
rtl/anvil_q1_mem_port.sv	49 total	49 code
rtl/anvil_q1_parameters.sv	27 total	27 code
rtl/anvil_q1_rmsnorm.sv	155 total	155 code
rtl/anvil_q1_rope.sv	97 total	97 code
rtl/anvil_q1_silu.sv	52 total	52 code
rtl/anvil_q1_swiglu_mlp.sv	75 total	75 code
rtl/anvil_q1_top.sv	151 total	151 code
rtl/anvil_q1_unsigned_divider.sv	45 total	45 code
rtl/anvil_q1_w4a16_gemv.sv	187 total	187 code

B Verification record index

Recorded check	Outcome	Execution time (s)
behavior	passed	615.587
sensitivity	passed	17.082
synthesis	passed	53.554
portability	passed	5.914
timing-repair-01-behavior	passed	637.655
timing-repair-01-sensitivity	passed	17.351
timing-repair-01-synthesis	passed	52.962
timing-repair-01-portability	passed	5.961
timing-repair-02-behavior	passed	644.465
timing-repair-02-sensitivity	passed	18.401
timing-repair-02-synthesis	passed	56.983
timing-repair-02-portability	passed	6.004
rmsnorm-sequential-01-behavior	passed	643.769
rmsnorm-sequential-01-sensitivity	passed	17.483
rmsnorm-sequential-01-synthesis	passed	54.057
rmsnorm-sequential-01-portability	passed	6.021

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E004	rtl/anvil_q1_decoder_layer.sv
E010	rtl/anvil_q1_rmsnorm.sv
E014	rtl/anvil_q1_top.sv
E015	rtl/anvil_q1_unsigned_divider.sv
E018	tests/official_weights.py
E035	.vulcan/engineering/run-0002/attempt-0996/stdout.log
E036	.vulcan/engineering/run-0002/attempt-0997/stdout.log
E037	.vulcan/engineering/run-0002/attempt-0998/stdout.log
E038	.vulcan/engineering/run-0002/attempt-0999/stdout.log
E040	.vulcan/engineering/run-0002/rtl-timing-lint.json
E041	.vulcan/engineering/run-0002/quartus-sta.json
E042	.vulcan/engineering/run-0002/board-hil.json
E043	evidence/verification/official_weights_comparison.json
E044	.vulcan/evidence/post-hil-q1/formal_24layer_dialogue.json
E045	.vulcan/evidence/post-hil-q1/multi_domain_dialogue.json

C Document provenance

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